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January 2, 1964

SUPERSEDING
Int. Fed. Spec. DDD-R-00250d (Army-GL)
December 17, 1962 and
Fed. Spec. DDD-R-250c
November 15, 1960

FEDERAL SPECIFICATION

RIBBON, ACCOUNTING MACHINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers black record ribbons that are suitable for use with electric accounting machines and high speed data processing printers (see 6.1).

1.2 **Classification.**

1.2.1 *Compositions and classes.* Ribbons shall be of the following compositions and classes, as specified (see 6.2).

Composition A.—Cotton.
- Class 1—320 thread count.
- Class 2—292 thread count.
- Class 3—270 thread count.

Composition B.—Silk.

Composition C.—Nylon.

1.2.2 *Degrees of inking.* Ribbons shall be furnished in the following degrees of inking, as specified (see 6.2):

Cotton ribbons:
- Medium inking.
- Heavy inking.

Silk ribbons:
- Medium inking.
- Medium-heavy inking.

Nylon ribbons:
- Light inking.
- Medium inking.

1.2.3 *Dimensions.* The lengths and widths of ribbons shall be as specified (see 3.5, 3.6, and 6.2).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:
- UU-P-121—Paper, Bond and Writing, White and Colored.
- CCC-T-191—Textile Test Methods.
- PPP-B-566 — Boxes, Folding, Paperboard.
- PPP-B-636—Boxes, Fiberboard.
- PPP-B-676 — Boxes, Set-Up, Paperboard.
- PPP-T-45—Tape, Gummed, Paper Reinforced.

Federal Standards:
- Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
- Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
- Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Weave. The fabric shall have a plain weave.

3.2 Edges. The edges shall be straight and parallel to the warp threads.

3.2.1 *Composition A, cotton.* The edges of cotton ribbons shall be woven or treated to prevent fraying.

3.2.2 *Composition B, silk.* The edges of silk ribbons shall be cut and treated to prevent fraying, or the edges shall be woven with a selvage, as specified (see 6.2).

3.2.3 *Composition C, nylon.* The edges of nylon ribbons shall be cut and fused.

3.3 Thread count. The thread count shall be the total of the warp and fill threads. Composition A ribbons shall be of the thread count as specified (see 1.2.1), with a tolerance of plus or minus 6 threads. Composition B ribbons shall have a minimum thread count of 266. Composition C ribbons shall have a minimum thread count of 274. In addition, composition C ribbons shall have a minimum of 164 warp threads and a minimum of 108 filling threads per square inch.

3.4 Thickness and breaking strength. Thickness and breaking strength of the ribbons shall conform to the requirements in table I when tested as specified in 4.4.1.1.

3.5 Width. The width shall be uniform. The composition A and B ribbons shall have a tolerance of plus or minus 1/64 of an inch. Composition C ribbons shall have a tolerance of plus or minus 1/32 of an inch, unless otherwise specified (see 6.2). Ribbons for Univac machines shall have a tolerance of plus 0, minus 1/16 of an inch. Ribbons of 7-1/2-inch width shall have a tolerance of plus or minus 1/16 of an inch.

3.6 Length. The length shall be as specified in table II (see 6.2). Ribbons shall be furnished in continuous lengths without joining or splicing.

TABLE I.—*Thickness and breaking strength*

Thickness (inch, maximum)			Breaking strength (lbs. minimum average)					
			Cotton		Silk		Nylon	
Cotton	Silk	Nylon	Warp	Fill	Warp	Fill	Warp	Fill
	0.0040	0.0050 ± 0.0005	—	—	63.5	61	100	70
Class 1—0.0045	—	—	38	42	—	—	—	—
Class 2—0.0050	—	—	43	50	—	—	—	—
Class 3—0.0057	—	—	55	58	—	—	—	—

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TABLE II.—*Length*

Specified length	Minimum acceptable length
200 yards ± 6 yards	194 yards
100 yards ± 3 yards	97 yards
55 yards ± 5 yards	50 yards
36 yards ± 6 inches	35 yards, 30 inches
20 yards ± 6 inches	19 yards, 30 inches
18 yards ± 6 inches	17 yards, 30 inches
16 yards ± 4 inches	15 yards, 32 inches
15 yards ± 4 inches	14 yards, 32 inches
12 yards ± 4 inches	11 yards, 32 inches
4 yards ± 2 inches	3 yards, 34 inches
3-2/3 yards ± 2 inches	3 yards, 22 inches
3-1/3 yards ± 2 inches	3 yards, 10 inches
2-2/3 yards ± 2 inches	2 yards, 22 inches

3.7 Color. Ribbons shall be black in color.

3.8 Writing qualities.

3.8.1 *Weardown.* When the weardown test is run as specified in 4.4.2.1, the first line shall be clear, clean, and easily legible to persons of normal or corrected 20/20 vision under an illumination of 25 foot-candles. The number of lines that are of good intensity, distinct, and easily legible shall be as shown in table III for the respective compositions, classes, and degrees of inking (see 4.4.2.3.3). There shall be no evidence of fabric damage upon completion of the weardown test.

3.8.2 *Recovery.* The ribbons shall show recovery (see 4.4.3) after making the weardown test. The intensity of the recovery line, in comparison with the lines that precede it, shall be no paler than the line number shown in table IV for the respective compositions, classes, and degrees of inking.

3.8.3 *Type filling.* There shall be no evidence of broadening or feathering of typed impressions when the ribbon is tested as specified in 4.4.4. The last small letter "e" shall be as legible and as sharp as the first "e".

3.8.4 *Oiliness.* There shall be no evidence of oil staining when the ribbon is tested as specified in 4.4.5.

3.8.5 *Permanence of writing.* Writing made with the ribbons shall show no marked fading when tested as specified in 4.4.6.

3.9 Performance. The nature of these ribbons and action that occurs in accounting machines and high speed printers are substantially different from laboratory equipment specified in tests. The supplier shall furnish satisfactory ribbons which will not spatter when tested under practical conditions, and the resultant impressions will not smear during normal handling.

TABLE III.—*Weardown requirements*

Degree of inking	Cotton—class			Silk	Nylon
	1	2	3		
Light inking	—	—	—	—	70
Medium inking	85	85	85	120	85
Heavy inking	120	120	120	—	—
Medium-heavy inking	—	—	—	150	—

TABLE IV.—*Recovery requirements*

Degree of inking	Cotton—class			Silk	Nylon
	1	2	3		
Light inking—no paler than line	—	—	—	—	55
Medium inking—no paler than line	70	70	70	110	70
Heavy inking—no paler than line	100	100	100	—	—
Medium-heavy inking—no paler than line	—	—	—	140	—

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3.10 Additional requirements used on IBM types 720, 730, and 1403.

3.10.1 *Compatibility.* The ink on the ribbon shall be compatible with paraffinic petroleum oil SAE 20. The ink shall readily mix with the oil, shall not flake or show signs of incompatibility when tested as specified in 4.4.7.1. Any oil deposited on the ribbon shall allow a satisfactory impression to be produced.

3.11 Shelf life. Ribbons in original unopened containers and not subjected to extremes of temperature and humidity, shall retain their serviceability as defined by this specification for not less than 12 months from date of receipt by the Government.

3.12 Spools, cores, retaining bars, and reversing attachments. Ribbons shall be attached on spools, cores, or retaining bars intended for use with the machine specified in the contract or order; and where applicable shall be provided with reversing attachments. Spools shall be of metal with an enamel or lacquer finish. Cores shall be of aluminum or resin impregnated paper or material suitable for the machine specified. When intended for use on machines requiring retaining or reversing bars, such bars shall be securely fastened to the fabric perpendicular to the outer edges of the ribbons. The distance between retaining and reversing bars shall be as required for the machine specified. When stitches are used, the stitching shall be uniform. Special features of the reversing attachments shall be as specified in the invitation for bids (see 6.2).

3.13 Workmanship. Ribbons shall be evenly woven and free from imperfections of manufacture which will affect their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 6.2.)

4.1.1 *Sampling for inspection and tests.* Sampling for inspection and tests shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all end items of the same class and composition.

4.2 Inspection of materials and components. The supplier is responsible for insuring that materials and components used in the end item were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.1 *Overall examination for visual defects.* Each defect listed below shall be counted not more than once for each spool examined. The sample unit for this examination shall be one spool. The sample size and acceptance number shall be as shown in table V.

Defects:

Objectionable odor.
Overall uncleanness.
Not composition, class, or degree of inking specified.
Color not black.
Weave not plain or evenly woven.
Ribbon edges not as specified.
Width not uniform.
Roll spliced or joined or not in one continuous length.
Spools, cores, retaining bars, and reversing attachments not suitable for machine specified.

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Spools not enameled or lacquered (applicable to electric accounting machines).
Stitching not uniform (when intended for use on machines requiring retaining and reversing bars).
Any tears, cuts, or holes in ribbon.

4.2.2 *Length and width examination.* Each spool in the sample shall be examined in accordance with CCC-T-191 for compliance with specified dimensions for length and width (see 3.5 and table II). Any dimension not within specified tolerance shall constitute a defect. The sample unit for this examination shall be one spool. The sample size and acceptance number shall be as shown in table V.

TABLE V.

Lot size in spools	Sample size in spools	Maximum number of defects acceptable in sample
5 and under ..	1	0
6 to 15	2	0
16 to 40	3	0
41 to 110	5	0
111 to 300	7	0
301 and over ..	10	1

4.2.3 *Examination of preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, contents, and weights comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L4 and the AQL shall be 4.0 expressed in defects per one hundred units.

Examine	Defects
Markings, exterior and interior	Incorrect; incomplete, illegible; omitted; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components; such as incomplete closure of case liners, container flaps, and loose or inadequate sealing, strapping or stapling.
	Bulged or distorted containers.
Contents	Number of units per intermediate package more or less than required.
	Number of intermediate packages per shipping container more or less than required.
Weights	Intermediate package exceeds 30 pounds.
	Contents exceed 65 pounds.

4.3 **Testing of the end item.** The methods of testing specified in CCC-T-191 wherever applicable and as listed in table VI shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3 continuous yards of finished ribbon. The sample size shall be in accordance with level II or MIL-STD-105. The lot shall be unacceptable if one or more units fail to meet any requirement specified.

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TABLE VI.—*Instructions for testing end item*

Characteristic	Specification reference		Number determinations per sample unit	Results reported as—	
	Requirement	Test method		Pass or fail	Numerically to nearest
Breaking strength	Table I	5100	Average of 5		Pound
Thread count	3.3	5050	Average of 5		whole number
Thickness	3.4	4.4.1.1	Average of 5		0.0001
Performance tests:					
Weardown	Table III	4.4.2	1	X	
Recovery	Table IV	4.4.3	1	X	
Type filling	3.8.3	4.4.4	1	X	
Oiliness	3.8.4	4.4.5	1	X	
Permanence of writing	3.8.5	4.4.6	1	X	
IBM types 720, 730, and 1403:					
Compatibility	3.10.1	4.4.7.1	1	X	

4.4 Test procedures.

4.4.1 *Atmospheric conditions.* Each sample ribbon shall be conditioned and tested at a temperature of 73° ± 3.5° F. and relative humidity of 50 ± 2 percent.

4.4.1.1 *Thickness.* Exceptions to method 5030 of CCC-T-191 shall be as follows: The dial of the gage shall be graduated to read directly to 0.0001 inch. The diameter of the presser foot shall be suitable for the width of the ribbon tested, so that the presser foot will not rest on treated, selvage, fused, or finished edges of the ribbon. When the width of the ribbon requires a presser foot smaller than the dimensions specified in method 5030 of CCC-T-191 the total load on the moving parts shall be decreased proportionately. The average thickness shall be reported to the nearest 0.0001 inch.

4.4.2 *Weardown (writing test).* The sample ribbon having satisfactorily passed the tests methods as specified in CCC-T-191 shall be subjected to writing tests.

4.4.2.1 *Equipment.*

a. Electric typewriter, IBM Model 11b, 11c or equivalent, pica type freshly cleaned (see 6.3). Number 1 platen (IBM) or platen with a durometer hardness of approximately 93 when measured with a Shore durometer, Model A (see 6.4).

b. Adjustment. The impression indicator shall be adjusted so that the impression force of the lower case letters will barely emboss the back side of one sheet of type III, 32-pound white bond writing paper conforming to UU-P-121. The multiple copy indicator shall be adjusted to the narrowest opening.

4.4.2.2 *Materials.* Samples from high-speed data-processing ribbons shall be cut with the filling threads in the long direction. Sample length shall be 5 inches. High-speed data-processing ribbons shall be cut to 5 inches by removing equal quantities from each side. Test sample width shall be no greater than 9/16 inch. Representative sample of the ribbon shall be cut with a minimum amount of handling to prevent the loss of ink.

4.4.2.3 *Preparation of samples for writing tests.*

4.4.2.3.1 *Accounting machine ribbons (see* 4.4.2.1). Unwind about 1 yard of the ribbon and lay the spool at one side of the machine with the end of the ribbon across the machine. Fasten the ribbon at opposite ends of the carriage under just enough tension (approximately 50 grams) to permit writing without slackness of the ribbon. The ribbon shall be adjusted at such height that the type will strike the middle of the ribbon.

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4.4.2.3.2 *Ribbons for high-speed processing printers.* The ribbon sample shall be secured to the carriage using suitable leaders. One set of leaders may be made by fastening the ribbon to the carriage using pressure sensitive tape. The junction between the ribbon and the leader can be secured with a staple. The portion of the tape tangent to the platen should be covered to prevent the tape from adhering to the platen. The ribbon shall be adjusted at such height that the type will strike the middle of the ribbon. The line guides of the typewriter shall be removed to permit free passage of the stapled portion.

4.4.2.3.3 *Preparation of impressions.* Insert a sheet of 32-pound white bond paper, conforming to the requirements for type III of UU-P-121, and long enough to complete this test. Then write a line of 30 capital R's. Push the carriage, bearing the ribbon with it, back to the starting point. Shift the paper a single-spaced line, and again write the same letters over the same portion of the ribbon. Repeat until the required number of lines have been written. It is essential that the entire number of lines be typed without interruption (see 3.8.1).

4.4.3 *Recovery.* Leave the ribbon in the machine for 1 hour for accounting machine ribbons and 5 minutes for high-speed data-processing printer ribbons. Then write another line of the R's over the same portion of the ribbon (see 3.8.2).

4.4.4 *Type filling.* Transfer the ribbon to a spool suitable for the testing machine, with as little handling as possible, and thread it through the machine in the usual manner. After thoroughly cleaning the small letter "e" make 800 impressions with it under the normal travel of the ribbon. The impressions shall be examined for evidence of filling (see 3.8.3). Ribbons for high-speed data-processing printers shall be cut in the warp direction. The center portion of the ribbon shall be obtained by removing equal amounts from each edge. The sample shall not be wider than 9/16 of an inch. Approximately 80 inches of ribbon are required for this test.

4.4.5 *Oiliness.* After 24 hours, the test sheet made as described in 4.4.2.3 shall be examined for evidence of oil stains around the letters. Both sides of the sheet shall be examined (see 3.8.4).

4.4.6 *Permanence of writing.* Portions of the test sheet made as specified in 4.4.2.1 shall be used for testing the permanence of writing. Lines 5 to 15 shall be cut from the test sheet and exposed to the radiation of a glass enclosed carbon arc lamp in accordance with method 5660 of CCC-T-191 for 48 hours. The writing shall show no marked fading. The development of a brown tone, if not accompanied by a definite fading of the writing, shall be disregarded. If there is any marked fading of the writing, lines 10 to 20 above the last line of the test sheet shall be exposed in the same manner. If the lines of the second set fade markedly, the ribbon shall be considered as not meeting the requirements of this specification (see 3.8.5).

4.4.7 *Additional tests for ribbons used on IBM types* 720, 730, *and* 1403.

4.4.7.1 *Compatibility.* A 1/16-inch piece of the sample ribbon shall be placed on a glass slide and 3 drops of paraffinic petroleum oil SAE 20, shall be placed on the ribbon. Observe for 3 to 5 minutes under approximately 25 power magnification.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2 and 6.5).

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5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Ribbons shall be individually wrapped in waxed paper, airtight cellophane or other suitable material, and enclosed in an individual metal, paperboard, fiberboard or other suitable container.

5.1.1.2 *Intermediate packaging.* Twelve ribbons of one composition, class, width, and length, unit packaged as specified in 5.1.1.1, shall be intermediately packaged in a snug-fitting paperboard box conforming to PPP-B-566 or PPP-B-676, except that weight limitations shall not be exceeded, or a snug-fitting fiberboard box conforming to type I or II, class 1, style RSC of PPP-B-636. Intermediate packaging shall not be required when 12 unit packages exceed 30 pounds. Each box shall be securely closed in accordance with the appendix to the box specification.

5.1.2 *Level C.* The ribbon shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Ribbon, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification. The weight of contents of each shipping container shall not exceed 75 pounds.

5.2.2 *Level B.* Twenty-four intermediate packages (288 ribbons), of one stock number, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. Each box shall be securely closed in accordance with the appendix to the box specification. Alternatively, boxes may be secured by the two strip tape method in accordance with Rule 41, Section 7 of the Uniform Freight Classification, using tape conforming to PPP-T-45. The weight of contents of each shipping container shall not exceed 75 pounds.

5.2.3 *Level C.* The ribbon, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 *Additional marking.* In addition to the marking specified in 5.3.1 and 5.3.2, unit, intermediate, and shipping containers shall be marked with date of pack (month and year).

6. NOTES

6.1 Intended use. Cotton accounting machine ribbons are intended for general purpose use and when there is a preference for heavy inking. Silk ribbons are intended for use when sharpness of writing and other performance characteristics of these ribbons are required. Nylon ribbons are intended for use with high-speed data-processing printers.

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6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Composition and class of ribbons (see 1.2.1).
(*c*) Degree of inking required (see 1.2.2).
(*d*) Whether edges of ribbons shall be treated or woven with selvage (see 3.2.2).
(*e*) Width and length of ribbons, manufacturer's make and model number of machine, and any special features of reversing attachments (see 3.5, 3.6, and 3.12).
(*f*) Selection of applicable levels of packaging and packing (see 5.1, 5.2, and 6.5).
(*g*) Any special marking for shipment required (see 5.3).

6.3 **IBM typewriter.** IBM Typewriter may be obtained from IBM Corporation, 590 Madison Avenue, New York, New York.

6.4 **Shore Durometer, Model A.** Shore Durometer, Model A may be obtained from Shore Instrument Manufacturing Company, Incorporated, Jamaica 35, New York.

6.5 **For civil agency procurements.** When level B packaging is specified, the requirements for level A shall apply.

MILITARY CUSTODIANS:

Army—GL

Navy—S&A

Air Force—MOA

PREPARING ACTIVITY:

Army—GL

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